



Interpol review of forensic firearm examination, 2022-2025

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1. Introduction

1.1. Scope

This review paper covers the publications concerning firearm examination, published from 2022 until and including 2024. A literature search was conducted covering articles on this topic published in the main forensic journals:

- AFTE Journal
- American Journal of Forensic Medicine and Pathology
- Australian Journal of Forensic Sciences
- Forensic Science International
- Forensic Science International: Synergy
- Forensic Sciences Research
- International Journal of Legal Medicine
- Journal of Forensic Identification
- Journal of Forensic Sciences
- Science and Justice

1.2. Current topics

The forensic firearm examination literature of 2022, 2023 and 2024 covers a range of topics. Within the topic of firearm examination the focus lies on validation and proficiency studies to provide insight on the validity of the discipline and in the performance of examiners. In addition, effort is spent on the development of computer-based methods, utilizing 3D measurements of the features in fired bullets and cartridge cases, to facilitate examiners to reach their judgments on the source of fired bullets and cartridge cases. Both these efforts are in line with the recommendations of the NRC report "Strengthening Forensic Science in the USA: A Path Forward" [1] and the PCAST report "Forensic Science in Criminal Courts: Ensuring Scientific Validity of Feature-Comparison Methods" [2].

When looking at papers published on shooting incident reconstruction we see that most effort is spent on gaining insight in the behavior of bullets and the performance of examiners when performing bullet path reconstructions with various methods.

Studies on wound ballistics mainly focus on the use of tissue simulators and the impact behavior and effect of bullets. Insight in the behavior of impacting bullets is not only beneficial to provide judgments on lethality, but could also affect shooting scene reconstructions.

The published articles related to forensic firearm examination are addressed in the sections "Firearm examination", "Firearms and ammunition miscellaneous reports", "Technical examination", "Shooting incident reconstruction", "Wound ballistics", and "Meetings" including their subsections on more specialized topics.

2. Firearm examination

A main task of firearm examiners is to provide judgments on the source of fired bullets and cartridge cases. To do so, they make use of the striations and impressions in these fired ammunition parts. These features result from imperfections in the various components of a firearm. To provide a judgment about the source of cartridge cases or bullets they take into account both the observed degree of similarity as well as their judgment about the characteristics value of the mark.

The evaluation process and reporting of comparison findings receive increasing scrutiny. Patteet and Champod have provided a concise overview of the criticism, applied methods and reporting formats in relation to criticism as put forward in the PCAST report [2]. They evaluate the pros and cons of the methods and reporting formats to try and enhance the credibility of the disciplines. They propose the use of a probabilistic framework and when possible the assistance of computer-based methods. They explain that this probabilistic approach overcomes issues that are inherent to the AFTE Theory of Identification in terms of lack of transparency in the justification of conclusions and decisions [3].

In addition to discussing the applied methods and reporting formats Patteet and Champod also discuss the used terminology. They propose to substitute the traditional terminology of class, subclass and individual characteristics by "due to design" and "accidental" [4]. They explain that this terminology fits better with the underlying mechanisms that resulted in the features and facilitates the probabilistic evaluation of observed degrees of similarity. They explain that the term subclass characteristics, which are hard to distinguish from individual

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characteristics without having access to knowledge about all production lines, could simply be suppressed.

2.1. Validation and foundational studies

Studies are performed to assess the characteristics value of specific marks. These are typically performed in a worst-case setting employing consecutively manufactured firearm components. These consecutively manufactured components provide the highest potential of similarity in features between firearms as a result of the manufacturing process. Overall speaking, examiners expect high inter-variability in features (low degrees of similarity) when shots are fired with two different firearm and low intra-variability (high degrees of similarity) when shots are fired with one firearm. Studies showing high inter-variability when using consecutively manufactured firearm components provide support for this general assumption through specifically tested firearm components. Studies showing low inter-variability would falsify this general assumption and could help examiners to recognize pitfalls of the assumption such as seen for subclass characteristics.

One such a study examined consecutively manufactured double-broached barrels of Ruger 9E pistols [5]. Barrel casts showed a lot of long and continuous striations, but also shifts and breaks in these striations over the complete length of the barrel. The author concludes that subclass carryover from barrel to barrel was limited and only apparent in single or double consecutively matching striations which are also not uncommon for known different-source comparisons. The author believes that each barrel produced unique and identifiable striations.

Consecutively manufactured barrels with the new rifling profile from Glock, the Glock Marksman Barrel, were examined [6]. Test shots and casts of ten barrels were compared. The author identified no evidence of subclass carryover on either the casts or the test fired bullets. No significant agreement in striations was observed between the barrels for the land engraved areas, groove engraved areas or channel/rail areas.

Ten consecutively manufactured breechface insert from Smith & Wesson's M&P Shield EZ M2.0 pistols were evaluated [7]. These removable insert are created by means of metal injection molding introducing the potential for subclass characteristics. While they all shared gross mold marks they also displayed random looking granular marks. Three examiners completed all comparisons in a partially open test based on these ten inserts and one additional non-consecutively manufactured insert. No misidentifications were reported. Based on the result, the author suggests that these inserts show individual markings.

Ten consecutively manufactured Thompson/Center Arms G2 Contender barrels were provided by Smith & Wesson. These barrels were cast after the reaming process (mid-production phase) and again after button rifling and crowning (completed production) [8]. Three barrels were sent back for secondary crowning and one barrel was sanded in increments of 1 inch from chamber to muzzle. Examination of casts and test fires showed that reaming marks remain in the barrel after button rifling. Secondary crowning drastically affected the resulting striations in bullets, which was attributed to the piloted crowning tool abrading the lands at the muzzle end of the bore. Sanding of imperfections near the muzzle of the bore had the most drastic effect on the striations on test fired bullets.

The results of a study involving thirty consecutively manufactured 9 mm Luger Beretta barrels [9] were reanalyzed [10]. In this study 72 examiners provided judgments on comparisons. Reanalysis was done in accordance with the PCAST's [2] suggestion to use maximum likelihood estimates to calculate false positive and false negative rates on only conclusive judgments. The reanalyzed false positive rate was 0.063%, 95% CI [0.03%, 0.14%]. When typos were corrected this dropped to a false positive rate of 0.009%, 95% CI [0.00%, 0.057%].

The false match probability when comparing firing pin shear marks of 9 mm Luger Glock pistols was investigated [11]. A sub-sample of 617 test fires of a total of 3156 test fires from as many different firearms was

intercompared using the Integrated Ballistic Identification System (IBIS). All 3156 test fires were intercompared by two examiners. No "matches" were reported by either methods and a false match probability of 0.4% was calculated.

2.2. Subclass characteristics

Broaching, plunge milling and lathe turning were studied for their potential to result in subclass characteristics [12]. The 3D topographies of consecutively manufacture breechfaces for a Thompson/Center Contender G2 pistol were compared using NIST's congruent matching cells algorithm. Based on the comparison result, the authors conclude that the presence of subclass characteristics was observed for broaching and plunge milling, but not for lathe turning. Finishing the surface with glass bead blasting reduced the effect of subclass to a higher degree than was seen for tumbling.

Otero reports on observing repeating milling patterns along the surface of the bolt face of a Mossberg 500 shotgun [13]. The author suggests that this repeating pattern show the presence of subclass characteristics in these firearms.

2.3. Class characteristics

Class characteristics can be used to assess if a cartridge case or bullet could have been fired with a firearm before examining the fine striations and impressions and to assess the make or model of firearm that could have been used to fire a cartridge.

Glock pistols are known to have clearly defined class characteristics of which the firing pin aperture is the most apparent. A tear drop shape is observed for Generation 5 pistols, while a rectangular shape is observed for the earlier generations. These class characteristics are not specific to Glock as they are also seen with Smith & Wesson pistols of the Sigma and M&P series. The ejector of Glock generations 4 and 5 seems to create a scoring mark on the cartridge sidewall at a consistent position. These marks were not observed for the other generations or makes and could thus help identify the used make of firearm [14].

Twelve types of .223 Remington ammunition, having various velocity and weight combinations, were test fired with two firearms. This was done to examine if discrepancies occurred between the widths of the land and groove impressions, and the class characteristics (general rifling characteristics) of the barrel. No inconsistencies were observed [15].

2.4. Examination based on specific markings

Ejection port marks can be found at the edge of the cartridge case of unfired cartridges and may also be observed on the bullet after ejection from a Glock pistol [16]. These marks result from ejecting the cartridge, which hits the lower left edge of the ejection port. These marks were observed on .380 Auto, 9 mm Luger, .40 S&W, 10 mm Auto and .45 Auto caliber cartridges. The author believes these marks to be specific to a firearm and could therefore be used to assess the source of the marks.

Some currently manufactured Glock pistols have an ejection assist groove in the corner of the breechface just over the ejector cut-out. This groove produces striations on the cartridge's rim or inner edge of the rim when cartridges are fired or cycled. Evaluation of the manufacturing process and resulting granular surface provides no indication for potential subclass characteristics [17].

While typically quite small in area, the comparison of features in ejector marks can be used to help establish the source of fired cartridge cases. Examples of such comparisons are provided by the author [18].

Cartridge cases fired from AK-type firearms often show a mark resulting from the dust cover (sometimes referred to as a "slap mark"). A Saiga 7.62 × 39 mm Rifle was used to examine the durability of the features in this mark. Test shots with 200 cartridges, of which 100 had a brass cartridge case and 100 a steel cartridge case, showed that both the

dust cover and resulting marks changed over time. This was especially apparent for the steel cartridge cases. The author notes that the marks became less distinctive over repeated firings, as the dust cover got somewhat polished [19].

2.5. Parameters that affect the identification process

A study examined the effect of decomposing human tissue on striation patterns in fired bullets [20]. Fired FMJ bullets were positioned in various places in and near human bodies. Over time the striations became less distinct and some of them became nearly indistinguishable. Some land impressions started to display pitting after prolonged exposure to decomposition. While the detrimental effects were most apparent as a result of the active decomposition stages, no clear effect of positioning in or near the body was observed. Based on their result, the authors advise to promptly remove any biological material from bullets when received for comparison purposes.

Another study on the effects of body decomposition on fired bullets was performed in Australia. Twenty-one copper jacketed, fired, bullets were inserted in lungs, abdomen and leg muscle (7 each) of human donors [21]. The effect on the bullets was studied over a time period of three weeks for both cool and warm conditions. The results showed that the bullets corroded quicker in warm conditions. The authors also emphasize the need to quickly remove any remaining tissue of bullets that are submitted for examination.

Subjecting bullets recovered from human bodies to reagents to remove tissue may affect the striations. Decalcification and formalin solutions where testes [22]. The authors state that no detrimental effect was observed when testing full metal jacket (copper) bullets. Degradation was observed after longer exposure times (2 to 3 weeks) for 9 mm Black Talon SXT and Silvertip bullets.

One option to create test fires when firearms are deemed unsafe to fire is the use of primed cartridge cases, where the bullet and propellant are removed. Visual comparison, using NIBIN, of the features in the firing pin impression, breechface impression and aperture shear marks between test shots of primed and normally loaded cartridges indicates no significant difference [23]. Only the breechface impression seemed less distinct. While the images were not correlated with NIBIN the author believes that comparison should be possible.

Modified blank-firing pistols are often encountered in some countries. These modifications can affect the resulting features in cartridges cases, which was tested for firearms modified to 7.65 mm Browning. The authors state that a higher degree of intra-variability was seen than for standard firearms. This will negatively affect the potential for algorithms to find candidate cartridge cases that could have been fired with the same firearm [24].

Certain ammunition-related parameters affect the transfer of marks from a firing pin to a primer [25]. The authors demonstrate that marks are produced in two stages: an impression stage and a second stage, when the pressure build-up in the primer pocket pushes the primer in the breechface and over the firing pin tip. This second stage is ammunition dependent.

2.6. Proficiency testing

The Houston Forensic Science Center (HFSC) reports on their blind proficiency test which ran from 2015 until 2021 [26]. The idea of blind proficiency tests (using mock or fake cases) is to test examiner performance in an actual casework setting while being unaware of being tested [27,28]. Fifty-one blind cases with a total of 570 examinations were reported by the HFSC. No misleading judgments were provided, but 40.3% comparisons resulted in an inconclusive judgment. Bullets proved the most significant factor predicting an inconclusive judgment, while variables such as experience, assigned examiner, intended complexity of the case and training program did not significantly contribute to these predictions. A blind proficiency program such as this demonstrates the

performance as applied in actual casework circumstances and can be tailored to address specific quality management aspects.

Test fires from ten consecutively manufactured Thompson/Center Arms G2 Contender barrels were used in a proficiency test [29]. Sixty-six firearm examiners from three countries, including 21 US states, enrolled of which 44 returned their results. The test kit consisted of 15 comparisons of which 5 questioned bullets were damaged by firing into drywall, glass, sheet metal, Kevlar and wood. Participants were allowed to use their standard operating procedures. The reported false positive (identification) rate was $0.455 \pm 0.139\%$ (1/220) and the false negative (elimination) rate was $1.82 \pm 0.555\%$ (8/440) at a 95% confidence interval. The total error rate for the 660 comparisons was $1.36 \pm 0.414\%$ (9/660).

Reproducibility of conclusions was tested with pairs of examiners with varying levels of expertise analyzing the same test sets [30]. A total of 15 test sets of fired bullets from the ten consecutively manufactured barrels of the previous study (see Ref. [29]) were compared by 20 examiners. In 77% (232/300) comparisons the same conclusion was given by two examiners. In 52 comparisons one examiner reported an identification and the other an inconclusive judgment, in 14 comparisons an elimination and inconclusive conclusion were reported and in 2 comparisons an identification and false elimination were reported by the paired examiners.

An error rate study involving 218 firearm examiners is published [31]. In this study each examiner performed 15 comparison sets consisting of one questioned cartridge cases and three knowns. The test was created based on test fires from 25 new Ruger SR9 9 mm Luger pistols. The false negative rate (false elimination) was 0.357% and the false positive rate (false identification) was 1.01%. A few of the participating examiners provided most of false positive judgments. An additional analysis on the use of the AFTE Range of Conclusions was performed on this study [32]. The authors state that the elimination category and the three inconclusive categories are not consistently used by the participants. They discuss patterns they observed, the potential causes and their implications.

A study involving 49 firearm examiner who performed 3156 bullet comparisons is reported [33]. The study consisted of bullets of varying quality (i.e., pristine to damaged), bullets from various makes and models of firearms, and various types of ammunition. Both one-on-one examinations (questioned vs questioned) and one-on-three examinations (questioned vs three knowns) were included. As the performance varied significantly over all conditions the authors believe it unwise to provide an overall error rate for the study. They do show that there was no significant difference in performance between one-on-one and one-on-three comparisons and that lower bullet quality resulted in a higher rate of inconclusive judgments which were especially common for bullets fired with polygonal barrels. Same-source comparisons between different types of ammunition showed a high rate of false exclusions.

The Collaborative Testing Service (CTS) firearms examination proficiency test 23-5262 was thoroughly examined and discussed by an AFTE review committee [34]. This was done after an unusual high number of false identifications were reported by CTS. The committee found no evidence of swabbed samples by CTS. No singular cause was found for the misidentifications. Contribution factors were discussed, such as 1) the believe that inconclusive judgments were no longer allowed or should not be given, 2) a high amount of time spent on the comparisons, potentially indicating difficulty of the comparisons, and 3) that lack of disagreement may have been conflated with significant/sufficient agreement. The committee states that errors were made and that the discussed factors are merely sources that could have contributed to them and should not be seen as justifications for the errors.

There has been quite some debate on how to assess error rates when "inconclusive" judgments are allowed in a study. Warren and Sheets propose an alternative performance analysis based on cross entropy that

can include inconclusive judgments [35].

In addition, Arora et al. [36] also propose a way to analyze reliability in black-box studies that apply an ordinal reporting scale and where examinations are repeated over time [36].

A group of 82 attorneys performed comparisons in order to compare the performance of examiners with that of laypeople to provide insight into expertise of examiners [37]. They were provided with images of the same comparisons as were previously compared by firearm examiners in earlier study [38]. The attorneys could generally distinguish between same-source and different-source comparisons. Their performance on different-source comparisons was almost comparable to that of firearm examiners while their accuracy rates were considerably lower for same-source comparisons. Commentary on this paper seems to focus on the critical attitude of the authors towards studied error rates and questions the scientific intentions of the authors [39] to which the authors respond from their perspective [40].

2.6.1. The FBI/Ames study

The planning, design and logistics of the FBI/Ames study are described [41]. The authors describe their study design aimed at assessing the performance of qualified examiners in the US. They will focus on accuracy, repeatability and reproducibility of examiner judgments on the source of fired bullets and cartridge cases. The authors state that the outcomes of the study will be presented in forthcoming publications. This approach of announcing a study and committing to publishing the results adds to a valid research culture and minimizes the potential for publication bias [28].

A total of 173 examiners participated in the announced FBI/Ames study and performed a total of 8640 bullet and cartridge case comparisons [42]. The overall false-positive error rates were 0.656%, 95% CI [0.305%, 1.42%] for bullets and 0.933, 95% CI [0.548%, 1.57%] for cartridge cases. The overall false-negative error rates were 2.87%, 95% CI [1.89%, 4.26%] for bullets and 1.87%, 95% CI [1.16%, 2.99%] for cartridge cases. Each examiner has a personal error probability, where a few examiners made most of the errors in this study.

Scurich and Stern [43] discuss that quite some of the eliminations of the FBI/Ames study were based in difference in class characteristics which seems to contradict the intended study design [41]. The researchers explain that this could be an unintended effect of the used Jimenez pistols of which the bolt inserts could have been produced by different machining process of different manufacturers [44]. The researchers emphasize that this does not imply that examiners are unclear about the definition of class characteristics.

Gutierrez and Addyman [45] discuss an alternative way of looking at the error rates of the examiners that provided false positive and/or false negative conclusions in the FBI/Ames study. They show that removing inconclusive judgments from the results greatly affects the error rates in a negative way. The overall false-positive rate for these examiners went up to 21.3% for bullet comparisons and 17.4% for cartridge case comparisons. Interestingly there was not much overlap between these two groups of examiners. Only 4 of the 24 examiners providing false-positive conclusion did so for both bullet and cartridge case comparisons. The correspondents imply that such results may raise the question if firearm examination should be admitted in court proceedings because of the lack of individual proficiency testing in practice. The researchers respond that the alternative way of calculating error rates indeed results in higher error rates but do not agree with the reasoning that the lack of individual proficiency testing should result in non-admissibility [46]. In addition, they emphasize that no verification took place in their study while this is common practice in casework. The absence of verification is assumed to have negatively affected the reported error rates.

In a follow-up study the repeatability (within examiners) and reproducibility (between examiners) of firearm examinations was studied by blindly resubmitting previously used comparison sets of the FBI/Ames study to examiners [47]. Taking into account all five levels of the AFTE conclusion range, an overall repeatability of 78.3% was found

for known matches and of 64.5% for known non-matches. The average reproducibility was found to be 67.3% for known matches and 36.5% for known non-matches.

The conclusion that the results of the previous study [47] shows consistency in judgments is discussed [48]. They believe that Cohen's kappa is far too low in this study to be able to conclude this. The researchers respond to this by explaining that they believe kappa is not a good measure of consistency for their study and that the commentator seems to discount the effect of comparison difficulty [49]. The researchers seem to assume a negative relationship between comparison difficulty and expected consistency.

In addition to their published findings on accuracy, repeatability and reproducibility, background information is provided on for example the participants' laboratory policies, standard operating procedures, training and experience, and documentation of judgments [50].

2.7. Development of computer-based methods

2.7.1. Virtual comparison microscopy

Traditional light comparison microscopy (LCM) is currently the main instrument used to compare features resulting from firearms. The use of virtual comparison microscopy (VCM), using 3D scans of the same features, is compared to LCM in terms of examiner performance [51]. Thirteen examiners completed fifty comparisons each using LCM for half of their comparisons and VCM for the other half. Results showed that correct positive identifications were made more frequently with VCM where LCM resulted in more inconclusive judgments. Both sensitivity and precision were higher for VCM (88.41% and 13.64%, respectively) than for LCM (80.08% and 12.50%, respectively). These results indicate that VCM is a valid technique to be implemented in casework and may even outperform LCM for complex comparisons.

2.7.2. Type and model determination

In an effort to determine the type and model of used firearms more objectively several convolutional neural networks were trained to differentiate between class characteristics of seven categories of firearms (Beretta, CZ, FN, Glock up to Gen 4, Glock Gen 5, Star, and Smith & Wesson). Prediction accuracy ranged from 71% to 81% and could be increased to 88% by combining machine learning models [52].

2.7.3. Bullet comparison

Evofinder is used for the evaluation of bullets fired with 1000 Norinco QSZ-92 pistols [53]. The similarity scores from the system were evaluated by calculating both ROC curves and score-based likelihood ratios. The authors explain that their results show an excellent discriminative power of the system and provide a summarized table to relate comparison scores to score-based likelihood ratios for more objective reporting on bullets fired with these firearms.

Four open-source approaches (cross correlation, congruent matching profile segments, consecutive matching striations, and a random forest model) to compare striations in bullets were evaluated using four datasets of bullets fired with consecutively manufactured barrels [54]. In addition, a Bayesian network was constructed that took into account the strengths of each separate model and proved to outperform the others when applied in isolations. The combined model classified 99.6% of the bullets correctly.

A method was developed to compare features in land engraved areas (LEA) of (air gun) bullets [55]. The researchers removed the bullets' curvature using loess fit and extracted features using Empirical Mode Decomposition and some entropy measures. They identified the features using minimum Redundancy Maximum Relevance. They performed classification of LEA using Decision Tree (DT), Support Vector Machine (SVM), and Random Forest (RF). In addition, they trained the DenseNet121 deep learning model to classify LEA. This deep learning model showed better performance than DT, SVM and RF. The authors suggest that the proposed method is helpful to examiners and include a

Grad-CAM technique to visualize the regions that were considered discriminative by the model.

Another method to compare the features in LEA uses finite ridgelet transform (FRIT) to extract striations and gray level co-occurrence matrix (GLCM) to generate linearly weighted average profile calculation [56]. By first dividing the surface in cells the valid correlation areas are assigned a high weight, suppressing those with invalid correlation areas that receive a low weight. A test with 35 bullets fired with 10 consecutively manufactured 9 mm Luger Ruger P85 barrels showed good separation between same-source and different-source comparisons and no incorrect identifications or exclusions were made.

2.7.4. Cartridge case comparison

One step towards a more automated approach of comparing features is automated region of interest (ROI) selection. U-net segmentation, deep learning, models were optimized and trained to determine the ROIs for breechface and firing pin impressions [57]. The authors believe this technique is promising for automated segmentation.

A method to extract features and to calculate a feature-based likelihood ratio instead of a score-based likelihood ratio was proposed [58]. The researchers collected 10 test fires from 300 firearms that all had a hemispherical firing pin and parallel breechface marks. Different approaches were tested of which the Zernike moment magnitude and phase features performed best. No (timewise) costly data rotation procedures were necessary and best performance was achieved when the whole region of interest (firing pin impression, flowback and breechface impression) was considered directly instead of first segmenting them. For casework settings using three test fires seems sufficient to attain a good specific-source, feature-based likelihood ratio.

A common source, score-based likelihood ratio method was developed for firing pin impressions, breechface impressions and aperture shear marks from Glock pistols [59]. Correlation coefficients combined with kernel density estimations are applied to calculate the LR for the striation patterns in the aperture shear marks and congruent matching cell (CMC) scores combined with parametric models for the firing pin and breechface impressions. Calculated LR are accompanied by an uncertainty measure and are limited in the score areas where little reference data is available. In addition, the authors demonstrate that the LR for the individual marks can be multiplied to interpret all marks as a whole.

A study utilizing nine Ruger model 10/22 rifles looked into a computer-based approach to compared firing pin impressions [60]. Several feature extraction algorithms were tested of which a computer vision algorithm (HOG – Histogram of Orientated Gradient) and a machine learning method (SVM – Support Vector Machines) showed the highest performance with an accuracy of 93%.

A Siamese network model, including two convolutional neural networks, was trained to compared firing pin impressions based on test fires from 300 casework firearms. The true positive rate at a prediction score of 0.5 was 0.87 and the true negative rate 0.88 [61].

A method to calculate likelihood ratios for granular breechface impressions was developed [62]. The method used the congruent matching cells (CMC) approach which was used to generate CMC scores for 130 same-source and 8385 different-source comparisons. These comparisons result from two test fires each from 130 firearms manufactured by eight manufacturers. Test results show that a CMC score of 2 represents an LR of approximately 1 while for a CMC score of 6 or higher the calculated LR is over a million.

Similarity judgments of sets of cartridge cases by laypeople were compared to comparison scores of a computer-based method [63]. The participants' similarity judgments were superior to the scores of the algorithm to differentiate between same-source and different-source comparisons. Compared to the algorithm, participants seemed to perceive as much or more similarity in comparisons with low-similarity known non-matches compared to high-similarity known non-matches. One caveat of the study is that the algorithm could only take into

account the breechface while participants could also consider for example the firing pin. The authors discuss that a hybrid human-machine approach could provide better discriminative power than one of the two separately.

2.8. Ballistic imaging database

In an effort to make searches in the National Integrated Ballistic Information Network (NIBIN) more efficient, the use of the new Unified Score is studied [64]. The San Bernardino County Sheriff's Department Scientific Investigation Division program showed that 99% of their NIBIN Leads have a Unified Score of 2 or higher. Using this score as a threshold for correlations will reduce time spend compared to the current threshold of viewing the top 30.

Ballistic imaging databases assist examiners to find potential links in open case files. The performance of BALISTIKA is studied with steel 7.62×39 mm cartridge cases [65]. Breechface, firing pin and ejector marks were considered as well as a combined evaluation criterium. Ejector marks showed to be the most distinct criterium to find same-source cartridge cases and correlations scores for cartridge cases of the same brand were higher than between brands of ammunition. Success rates also seemed to vary between firearms from different origins.

The effects of implementing Ballistics IQ at the Institute of Forensic Sciences of Puerto Rico as an automated triage solution for NIBIN sites is discussed [66].

2.9. Unusual comparisons

Bolton examined the firing pin impressions in 21 mm Rocket Practice M72A6 Sub-Calibre munitions after firing with rocket launchers. The author concludes that the same comparison techniques can be utilized as are used for firearm examination to differentiate between sources [67].

While they are no firearms, nail guns can be used to cause harm. In a case, a fired nail was linked to the strip of nails in the nail gun by means of a physical-fit or souche of the paper tape that holds the nails together [68].

3. Firearms and ammunition miscellaneous reports

3.1. Firearms

3.1.1. Alternative firing mechanism

The Remington 700 EtronX is described [69]. This bolt-action rifle operates on an electronic ignition system in combination with a specific type of EtronX ammunition with an electric primer ignition system. The idea is that the replacement of a mechanical trigger and percussive primer will enhance accuracy.

3.1.2. 3D-printed firearms and components

As a result of technological advancements, 3D-printed firearms can be manufactured beyond governmental control. An analysis of media and law enforcement reports ($N = 186$) shows an increase in encounters with these types of firearm, especially since 2021. These data include seizures, online sales, and illicit use. 3D printing is most often related to hybrid models, parts kit completions or conversions and specific components such as sears for automatic firing [70].

Data collected between 2021 and 2023 of over 2100 3D-printed firearm plans show that blueprints of complete firearms were initially dominant online [71]. Nowadays parts kit completion or conversions (PKC) and hybrid designs have become more popular. The authors hypothesize that this is a results of their improved reliability and performance.

Examination of 3D-printed and hybrid firearms can be aided by micro-CT scans. These scans allow an assessment of the firearms' components, viability and mechanics before test firing and dismantling. The resulting digital twin ensures that valuable information is recorded in

case of damage during test firing and can help present and communicate evidence visually [72].

Six 3D-printed firearms (PM522 Washbear, PM422 Songbird, TESSA, TREVOR, Grizzly and Marvel Revolver) were used to assess their firing potential [73]. All six firearms were functional, but could only be used to fire one shot due to the resulting damage of the firearm. Firing resulted in ruptures, projected parts and fragments. Traces of melted polymer could be found on the fired ammunition as well as tears and swellings on the cartridge cases.

A Liberator was 3D-printed using PLA and ABS filaments and fired. The Liberator fractured to such an extent that only one shot was possible. No comparable features were observed. Potential traces of thermoplastic deposits were observed on both the cartridge case and bullet [74].

Chemical analysis of polymers in 3D-printed firearms can provide valuable information about their source. The application of FTIR and Raman spectroscopy of collected samples showed that polymer filaments have a low intra-variability, but a high inter-variability [75]. The high inter-variability is explained to be a result of different polymer types and used color pigments. This difference between intra- and inter-variability results in a high discriminative power using these techniques.

In addition to chemical analysis of the polymers used in 3D-printed firearms, the potential of using the visual layering and surface characteristics, that results from manufacturing, to provide a judgment about the source of components is discussed [76].

3.1.3. Image and video analysis

Establishing whether an object is a lethal firearm is typically performed by analyzing the object itself. Chaikovsky et al. (2023) propose that this might, to some extent, also be possible by photographic content analysis [77]. They state that by considering the esthetic, but more specifically the functional morphological features, and comparing those with video database of firearms and replicas this technique may be used to ascertain whether an object is a firearm. Event sequence reconstruction of the shooting process in videos could help with this assessment.

3.1.4. Hidden serial numbers

Hi-Point has a history of trying to assist forensic firearm examiners by designing their firearms with Hi-Point-specific rifling, sanded breechfaces and the use of hidden serial numbers [78]. For their carbines the location of the hidden serial numbers recently changed from the magazine well to the interior of the trigger plate. The trigger plate needs to be removed in order to observe the serial number.

3.1.5. Serial number and text restoration

Chemical etching reagents are often used to restore obliterated serial numbers. The reagents exploit the residual stress in the metal surface that results from cold pressing the serial number in the surface. Several etching reagents were tested for medium carbon (0.40% C) steel of which the authors determined that the standard Fry's Reagent produced the best results [79].

While chemical etching is an established method to restore serial numbers, the (partly) restored characters are not always easy to distinguish. A study involving 39 participants and 64 steel plates with six randomized, stamped, removed and restored characters showed that the letters B and N were most often misidentified. The letter G, Q and S also showed a noticeable increase in misidentifications [80].

A study with a similar design of steel plates (N = 118) with restored characters was performed using both examiners and novices to test their accuracy. Similar character determination accuracy was observed with error rates of 2.3% for examiners and 2.4% for novices. A difference was observed in the shown confidence, where examiners showed a more conservative nature [81].

Serial number recovery of laser engraved marks on steel was studied using Regula Forensics Eddy Current Recovery followed by Fry's reagent

[82]. Zero out of the 509 laser engraved and obliterated marks were recovered with this technique. The authors recommend manufacturers to avoid the use of laser engraving.

Two non-ferrous cylindrical metal pieces were examined as they were suspected to form a homemade firearm. Partial texts were visible which appeared to have been stenciled or painted. Fluorescent lights and the Crime Scope model CS-16-400 alternative light system at various wavelengths were utilized to visualize the text [83].

3.2. Ammunition

3.2.1. Manufacturing marks

Examination of eighteen 9 mm Luger Turan Ammo cartridges showed the presence of striated impressions on the primers [84]. Comparison of these features showed six groups of different tool origins. The author hypothesizes that these features could be the result of the primer seating process during manufacturing and provides a warning not to mistake these for firearm features if they would still be visible after firing.

3.2.2. Cleaning

Prolonged exposure to environmental conditions can lead to the deposition of organic material and corrosion of ammunition. A cleaning sequence was tested on WWII brass cartridge cases. In the sequence the cartridge cases were sequentially treated in ultrasonic baths with a diluted soft metal cleaner (Tickopur TR 7), sulfuric acid, and Aqua Regia. The authors state that this resulted in effective cleaning after which marks resulting from firearms could be compared [85].

3.2.3. Propellants

The potential to use size- and shape-dependent metrics to assess the brand of smokeless small arm propellants was assessed [86]. Size-dependent metrics seem to provide the highest discriminative power. The overall accuracy of the methods was approximately 85% at brand level. The method described could be of value for intelligence purposes or at the initial stages of investigations.

3.2.4. Trace analysis

The wad of a fired shotgun cartridge can potentially provide information about the used ammunition if the cartridge case itself is not recovered. Fired and non-fired (pristine) wads of several brands of ammunition were analyzed with infrared spectroscopy, X-ray diffraction and thermal analysis [87]. Based on these analyses the possible sample pairs in this exploratory study could be paired correctly. This shows that the proposed method is robust to the potential effects of pressure and temperature on the polymer structure of the wads.

When the absence of striation patterns makes it impossible to assess the source of a fired bullet looking into the lead (Pb) isotope signature may prove valuable [88]. Pb isotope ratios were measured by LA-MC-ICPMS. This was done for the fragment collected from a victim, for seized ammunition from two suspects, and for a reference set of 50 lead bullets from various brands. Likelihood ratios were calculated to quantify the degree of support for the considered source hypotheses.

Another study also looked into the application of MC-ICPMS for lead isotopes as well as ICPMS for trace analysis [89]. The authors conclude that looking at the lead isotopes provides a higher discriminative power than trace analysis with ICPMS. They do not apply this technique to associate a crime scene bullet with unfired ammunition, but apply it to support associations between for example a fired bullet and a ricochet mark or between bullets and their bullet fragments.

3.2.5. Special ammunition

Bullets with a lead core and polymer coating are available as Syntech projectiles under the American Eagle brand of Federal Premium Ammunition. The polymer coating is designed to decrease lead and copper fouling. While class characteristics may remain, this is not true

for the striation patterns [90,91].

Ruger ARX frangible self-defense ammunition was examined [92]. These injection-molded bullets with a polymer-copper matrix show the barrel's class characteristics, consistent with those seen in traditional bullets. Fine striation patterns did not carry-over however. The bullets seem to produce hexagon-like shapes when perforating sheet metal.

Information is provided about the manufacturing process of G2 Research Ammunition who are known for their R.I.P. and Civil Duty bullets [93].

Federal BallisticClean ammunition is a unique, lead-free, frangible projectile. It consist of a zinc wire wrap covered by a copper jacket. Pieces of zinc wire might be found in bodies by forensic pathologists [94].

3.3. Recovery of traces from firearms and ammunition

3.3.1. DNA

There are different techniques available for the recovery of DNA from fired cartridge cases. Recovery rates after swabbing, tape lifting and soaking were compared in a study using brass .22 Long Rifle and 9 mm Luger cartridges [95]. Observed recovery rates were significantly lower for swabbing than for tape lifting and soaking. No significant difference was found for the latter two, but tape lifting resulting in a higher DNA yield on average.

The potential value of direct PCR from 12 gauge shotgun shells was compared to DNA extraction using traditional swabbing. Recovery of alleles was significantly lower for direct PCR. The authors discuss that this could be the result of PCR inhibitors which may vary for different types of ammunition [96].

The DNA recover rate for different firearm components was studied based on New South Wales's firearm casework data from 2025 to 2019 [97]. DNA yields were lowest from the firearms' hammer and safety and from cartridges, cartridge cases and bullets. The highest DNA recovery was found for single swabs of multiple firearm components. Internal parts of a firearm more often resulted in profiles from one donor.

Similarly, DNA recovery from 617 firearm-related submissions from cases between 2020 and 2022 of the Palm Beach County Sheriff's Office Crime Laboratory Forensic Biology Unit were analyzed [98]. Suitable DNA profiles were seen in 53% of the samples where DNA was recovered. The authors provide an overview of successful and unsuccessful DNA recoveries from firearm components and firearm-related accessories.

3.3.2. Fingermarks

Various factors can affect the likelihood of obtaining fingermarks from fired cartridge cases. Three studies looked into this.

A fingermark recovery study was performed using the Foster + Freeman RECOVER Latent Fingerprint Technology, that utilizes disulfur dinitride (S_2N_2) [99]. Cartridge cases were fired after various intervals of fingermark deposition. Overall 9.3% of deposited fingermarks in this lab study were deemed identifiable post-firing. In a pseudo-operational study only 1.8% of the comparable fingermarks were developed, showing a large difference between a controlled lab study and casework.

With the aid of gas chromatography coupled with mass spectroscopy it was shown that all lipids that are detected in unfired cartridges can also be present in fired cartridge cases [100]. Apart from squalene, which was found to have a 28% lower recovery in fired cartridge cases, no significant concentration differences were observed for lipids pre- and post-firing. The authors discuss that this reduces the likelihood that temperature is the main factor that affects the loss of fingermark quality between pre- and post-firing.

Another study looking into the quality if developed fingermarks took into account both the time after treatment with Recover Latent Fingerprint Technology and fingermark deposition before firing [101]. They found an increase in quality in 18.3 to 29.2% of fingermarks after 48 h up to 7 days after treatment. When fingermarks were deposited 24 h

before firing 12.5% were deemed suitable for identification, while this dropped to 3.3% when they were deposited 1 h before firing.

3.4. Statistics

Firearm-related mass murders at schools, colleges and universities were studied based on data in the Columbia Mass Murder Database. Based on 82 mass murder incidents the authors show that 63.2% of these cases that involve firearms occurred in the US. Perpetrators were mostly Caucasian (66.7%) and male (100%). The majority of the perpetrators (80.7%) did not show any severe mental illness. And about half of the incidents end with the shooter taking their own life [102].

The claim that a decreasing trend in the firearm-related deaths in Durban, South Africa, could be ascribed to increased intelligence and policing [103] is discussed [104]. Very little change is observed from 2010 onwards, with an increase after 2012. This increase is said to be attributed to increasing firearm availability in the region.

The effectiveness of the Phoenix Police Department's Crime Gun Intelligence Center was assessed [105]. Comparison of data of a year prior to its implementation and two years after showed that NIBIN inputs increased with 115%, the number of NIBIN leads by 163%, and that 32% of items were processed within 24-48 h instead of the earlier 3%. In addition, a significant increase in clearance rates for arrests was observed.

Social network analysis can help show the use of firearms in multiple incidents and associated people, and to identify integral people in gun violence networks. Adding NIBIN leads and associated incidents in combination with incident reports and co-arrest data seems to provide more and valuable information that directly relates to gun crimes [106].

The evaluation of used ammunition in cases could support police investigations and provide intelligence on sources of ammunition. Data from four countries (Denmark, Germany, Sweden and Switzerland) from 2015 to 2020 showed that most ammunition was recovered from violent crimes and that a small proportion of the encountered 610 headstamps accounted for most cases. Trends such as the use of blank-firing ammunition in crimes in all four countries and the rise of 7.62×39 mm ammunition in one country were observed [107].

3.5. Human factor

Comparing features and providing judgments about the source of pieces of evidence is a main part of forensic firearm examiners' tasks. Studies on underlying human comparison expertise could provide insight in how examiners provide these judgments and which factors may impact effective training and accuracy.

Two studies with novices explored whether there is a generalizable and domain-general perceptual skill for this comparison ability [108]. Individual differences were compared in four domains, namely firearms, fingerprints, faces and artificial prints. Individual accuracy was correlated between all comparison tasks. This provides evidence for a domain-general visual comparison ability, which differs between people.

In another study the comparison performance of forensic examiners (firearms, fingerprints and faces) is compared to that of novices [109]. All participants performed comparison tasks of various disciplines (firearms, fingerprints, faces, and novel objects). The results show that examiners outperform novices and examiners from other disciplines within their domain of expertise. In addition, examiners seem to possess some form of domain-general visual comparison ability as they also outperform novices in other disciplines.

3.6. Reporting

A survey amongst 79 firearm and toolmark examiners provides some statistics on casework. The modal time spend on cartridge case comparisons is 1-2 h and 2-4 for bullet comparisons. On average participant

provide an identification judgment in 65% of casework, an elimination in 12% and an inconclusive in 20%. The authors also discuss that lab policies may affect the judgments of examiners and that there are mixed feelings regarding the introduction of statistics and technological approaches [110].

3.7. Court rulings

The Supreme Court of Maryland, US has filed a ruling in June 20, 2023. This ruling limits the submission of firearms evidence in hearings. The Supreme court ruled that firearm examiners cannot reliably support a categorical or unqualified conclusion that bullets or cartridge cases were fired with a specific firearm. The authors provide insight in the court's ruling, potential effect for the admissibility of firearm evidence and what is and is not allowed to be stated, and resulting discussions amongst examiners [111].

3.8. News from Australia and New Zealand

A paper provided an overview of 22 unpublished reports of studies [112]. The authors note that these studies are individually unsuitable for publication, due to several limitations, but might be useful to shape future research projects. An overview of the topics, result summaries and how the studies might be improved is presented.

4. Technical examination

4.1. Modified and homemade firearms

A Harrington & Richardson brand starter pistol was modified. Parts had been removed, altered, and added to convert it to a .22 Short caliber revolver which could fire potentially lethal projectiles [113].

Examination of a "Colt" double action revolver showed an overly long cylinder engraved with the numbers "223". Examined features led to the believe that this was a forgery [114].

A counterfeit Jericho 941 RPSL was examined [115]. Although all markings indicated it was an authentic Jericho pistol, some aspects seemed off. In-depth examination of the pistol, including overwritten markings, led to the believe that this was a pistol produced by Jordan Armament & Weapon Systems that had been made to look like a Jericho after manufacturing.

The modifications and changes to the mechanical function are described for a Chinese Kalshnikov type 56 assault rifle that was modified from firing 7.62×39 mm ammunition to 5.56×45 mm ammunition [116].

Blank-firing pistols can be converted to pistols that are able to fire cartridges with projectiles such as bullets and steel balls. Methods of conversion are discussed as well as the class characteristics and potential for subclass of some models [117]. In addition, the authors show some pistol component failures resulting from firing after conversion.

4.2. Accidental discharge

A technical examination of a Stag Arms model 15 5.56×45 mm rifle showed that it had a broken hammer pin [118]. The hammer and sear were misaligned as a result. This caused the disconnector to malfunction. The effect of this was that releasing the trigger would result in another (accidental) shot after firing and in automatic mode. This corresponded with the officer's statement of firing two to three times, while six cartridges were missing from the magazine.

Three accidental discharges due to dropped firearms were reported [119]. In one case with an older Beretta model 70 semi-automatic pistol, the cause was found to be wear of vital parts. In the other two cases, involving a derringer type pistol and a Norinco model 54 semi-automatic pistol, a design flaw was found to be the cause. With both firearms it was possible to rest the external hammer on the back of the firing pin which

in turn rested with its tip on the primer of the cartridge in the chamber. When the firearm is dropped on the back of the hammer in loaded condition, a shot is fired.

4.3. Methods

Trigger pull measurements are often performed in cases where it is questioned if a shot was fired intentionally or accidental. For these measurements it is good practice to also provide a statement concerning measurement uncertainty. When establishing this uncertainty using a spring gauge, the Bexar County Criminal Investigation Laboratory found a high amount of variability between measurements. In addition, they noticed a significant difference in trigger pull between taking it at the central position of the trigger or the tip of the trigger [120]. Mean trigger pull was lower at the tip of the trigger for most firearms tested.

5. Shooting incident reconstruction

5.1. Research

5.1.1. Bullet behavior and bullet trajectory reconstruction

Various parameters affect the accuracy and reliability of a reconstructed bullet trajectory. The most well-known are: 1) the angle of incidence on impact, 2) the properties of the target material, and 3) the properties of the bullet upon impact [121,122]. Another parameter, the shooting distance, is mostly neglected for shorter shooting distances [123,124]. Gravitational pull causes a systematic error on the bullet trajectory's vertical angle. The magnitude of this effect varies for different firearm and ammunition combinations. Based on a study looking into bullet drop, drop angle and vertical offset, shooting distances of 20 m for subsonic/transonic handgun bullets and 30 m for supersonic handgun bullets are proposed as conservative shooting distances for which bullet trajectories can still be considered as straight lines. This will affect bullet path reconstructions by a bullet drop of no more than 5 cm or a vertical angle error of 0.3° [125].

Two studies focus on ricochet behavior of bullets on glazed ceramic tiles, often encountered in urban environments. For 7.62×39 mm bullets the reported critical ricochet angles, at which angle of incidence 50% of the bullets will ricochet, were found to be 16.0° for wall tiles and 10.0° for floor tiles [126]. Furthermore, the impact defects resulting from angles of incidence over 9° were visually similar to those of orthogonal shots. For 9 mm Luger bullets the observed critical ricochet angles were 16.6° for wall tiles and 14.8° for floor tiles [127]. In addition, the authors discuss the novel "caterpillar" ricochet mark they observed, showing curved semi-circular crack patterns which are believed to be stress fractures in the glazed surface of the tiles. The authors of both papers emphasize the need for case-specific empirical studies.

A study on bullet ricochet behavior on 1 mm sheet metal showed critical ricochet angles of 15.9° for 9 mm Luger bullets fired with a Browning HP, 15.0° for 9 mm Luger bullets fired with an Uzi SMG, and 11.4° for 7.62×51 mm bullets fired with an SLR rifle [128]. No critical ricochet angle could be determined for the 5.56×45 mm bullets as they fragmented upon impact. For the 7.62×51 mm bullets, a significant negative linear correlation was observed between the length of the ricochet mark and the angle of incidence. This relationship can help establish the angle of incidence from the resulting bullet defect.

Post-ricochet bullet behavior was studied for 7.62×39 mm bullets fired at various types of wood and tile samples [129]. Fairly consistent bullet orientations, with a rightwards jaw, were observed at a fixed, close-range, post-ricochet distance of 16.4 – 17.0 cm for wood and tile surfaces. For wood surfaces the jaw angles ranged from 25° to 72° , and for floor tile surfaces from 34° to 51° . For the used wall tiles, with a more irregular surface, results were less consistent with angles ranging from 33° to 147° .

Post-ricochet behavior was also studied for nine-ball shotgun pellets

of 8.6 mm diameter on standard floor tiles [130]. The authors demonstrate a significant relationship between the angle of incidence and the length and width of individual pellet defects. Furthermore, a positive relationship was found between the width of shot spread on tiles and on the witness screens.

The accuracy of reconstructed bullet trajectories from bullet defect on sheet metal (car, with a thickness of 0.85 mm) applying the lead-in method was studied for a total of 30 shots with one shot at each of the ten known angles per caliber (.22 LR, .45 Auto, and 9 mm Luger) [131]. Based on the results of three blind and two non-blind participants the authors show that each caliber has its own error curve, but that overall the error of the reconstructed trajectory was lower for lower angles of incidence (the observed error ranged from 1° to 13.9°).

A novel approach to assess bullet type from the resulting bullet defects in concrete was studied, applying a random forest machine learning algorithm [132]. Approximately 30 shots were fired with each of the five types of bullets (.22 LR, 9 mm Luger FMJ, 9 mm Luger Hollow Point, .45 Auto FMJ, and .45 Auto Hollow Point) at two angles of incidence (approximately 10° and 20°). Morphological dimensions (width and length) of the bullet defects and bullet defect perimeter-to-area ratio proved to be the top predictor variables for bullet type, with .22LR being the easiest to classify. While classification accuracy was shown to be over random classification of 20% and varied between 62% (without considering the angle of incidence in the model) and 66% (including the angle of incidence in the model) quite some overlap was seen between bullet types.

When shooting through laminated glass, as used for windscreens in cars, conically expelled fine glass spray is typically observed. In addition, a round-shaped compounded glass fragment can be found. This consists of several miniscule glass fragments held together by the PVC binding in laminated glass. These were observed after shots with 9 mm Luger FMJ and PEP 2.0 (MEN) bullets through laminated glass [133, 134]. The morphology of these round-shaped compounded fragments could be used to differentiate between the two bullet types. The wounding potential of these compounded fragments is higher than of the typical spray of fine glass fragments and was shown to penetrate ballistic gelatine up to 2.4 cm after perforating a T-shirt.

5.1.2. Shooting distance determination

While shooting distance determination is often based on GSR dispersion patterns there are also situation where the firearms themselves or the pellets from shotguns can be used to provide information about the shooting distance.

Visual staining, DNA recovery and RNA recovery were studied for 81 shots fired at reference cubes (12 cm length cubes of 10% gelatine with an embedded paint-blood-pad) [135]. Handguns of the calibers 9 mm Luger, .38 Special and .32 Auto were used and shots were fired with contact (0 cm) and at ranges of 1 cm up until 50 cm. In general, staining seemed to decrease for increasing shooting distances as well as the recovery of DNA and RNA. But a high degree of variability was observed within and between shots at the various shooting distances. Based on these results, the authors conclude that shooting distance determination based on DNA recovery only is not advisable.

Schyma and Berthold propose a technique using 11% gelatine and 1% alginate to visualize and preserve biological traces from the inside of barrels. They state that the technique proved effective in capturing blood and gunshot residue patterns in barrels of handguns [136].

Grisham demonstrated the use of the equivalent circle circumference method for shotgun distance determination [137]. To do so, he fired three types of loads at various shooting distances with a Remington 870 Express Magnum pump shotgun. The circumference was measured by summing the distance between the pellets at the periphery of the pattern. A strong linear relationship was observed between the shooting distance and the circumference of the shotgun pellets' pattern.

Important parameters when determining shooting distance based on shotgun dispersion patterns are the used firearm and ammunition. If

possible, the firearm and ammunition used in the incident should be used for test shooting. Often the correct ammunition is not available in sufficient quantity to perform all necessary test shots. In such a situation, opting for ammunition of the same brand with a pellet size of the next nearest shot number may be a valid option. A study on the relationship between pellet size and shotgun dispersion patterns provided support that shotgun pellet dispersion increases with decreasing pellet size for both steel and lead shot [138]. Although the magnitude of this effect was clear for large difference in pellet size it was not as clear for consecutive shot sizes. Differences in shot patterns were fairly small.

The angle of incidence seems to affect the amount and density of GSR on the target material [139]. When the angle of incidence decreases, the amount and density of GSR decrease as well. This is an important factor to consider when performing shooting distance determinations. In addition, elongated GSR patterns are observed for non-orthogonal shots of which the width seems to be decrease with decreasing angle of incidence.

The used propellant in a specific commercially available cartridge may vary over the months or years of production. Changes in used propellant that provide the same muzzle velocity and peak pressure may affect gunshot residue patterns which could lead to errors in shooting distance determinations [140].

The accuracy of shooting distance determination for TASERs is studied [141]. The impact spread between the two barbed probes is positively related to shooting distance. Standard deviation in calculating the shooting distance increased with distance. The authors provide standard errors for the shooting distance estimations for the X2, TASER 7 CQ, X26P and TASER 7 Standoff model.

5.2. Methods

3D laser scanners are increasingly used to record crime scenes, including reconstructed bullet trajectories. When reporting the results from any shooting incident reconstruction it is important to consider both systematic error (between measurement and ground truth) as well as reliability or measurement uncertainty [122]. This measurement uncertainty was studied by the Washington State Patrol Crime Scene Response Team who applied Trimble X7 3D laser scanners [142]. They looked into 1) the intra- and inter-variation following shots on flat surfaces, 2) the intra- and inter-variation following shots on curved surfaces, and 3) accuracy when compared to one shot of which the vertical and horizontal angle were known. Based on these studies they report an overall measurement uncertainty of $\pm 2.6^\circ$ at a 95% confidence interval for trajectory reconstructions based on rod placement.

The Light Detection and Ranging (LiDAR) sensor in the more recent iPhones and iPads can also be used as a low-cost alternative for terrestrial 3D laser scanners. Their applicability was tested in combination with the iOS application Recon-3D and the results were compared to those of the FARO Focus S350 [143]. Tests were performed at various angles in incidence in both a controlled scenario (lab study with shots fired at a piece of MDF) and a practical scenario (outside study with shots fired at a white car). While their results may only relate to the tested conditions, the authors report that angle errors did not significantly differ between the FARO focus S350 and the Recon-3D. If these results prove to be robust over different conditions, this could potentially be a low-cost and handheld alternative to terrestrial 3D laser scanners.

Bullet deformation can provide valuable information concerning the bullet's trajectory. To study the expected bullet deformation given specific scenarios it is important to perform empirical studies on bullet deformation. To collect bullets mid-flight, with non to minimal additional deformation, a cost effective bullet recovery device consisting of stratifications of Kevlar and cardboard is proposed. This method seems to work well for jacketed handgun bullets [144].

5.3. Case reports

Levy et al. (2022) report that standard operating procedures usually focus on the analysis of outer layers of garment to assess shooting distance. In their case they decided to also examine the inner layers, where they found traces of GSR. The abundance and distribution of GSR led to a new alternative scenario of what might have happened. The authors emphasize the need for thorough inspection of forensic evidence [145].

6. Wound ballistics

6.1. Research

6.1.1. Tissue simulants

Ordnance gelatine, prepared with a 10-90% (by weight) gelatine to water ratio and cooled to 4 °C is the standard soft tissue simulant in wound ballistics. This simulant has several disadvantages, one of which is that it is temperature sensitive. A new study describes how adjusting the temperature can be used to bring gelatine blocks within calibration [146]. For example, when a block is slightly out of calibration because of BB overpenetration, the firmness of the block can be increased by lowering its temperature. The study provides the relationship between gelatine temperature and BB penetration depth, to be able to select the right temperature.

Ordnance gelatine was compared to synthetic alternatives in another study [147]. There was no significant difference between the energy transferred to 40 cm long blocks for of ordnance gelatine (10% and 20%) and three types of synthetic products (Clear Ballistics and Perma-Gel). There was a significant difference between ordnance gelatine and the synthetic products when local energy transfer was calculated based on the fissure lengths. However, there was no significant difference when local energy transfer was assessed using high-speed video as a tool. This suggests that all simulants respond in a similar manner, at least visually.

A natural (cowhide) skin simulant was bonded to ordnance gelatine to produce a combined skin/soft tissue simulant [148]. This practice opens ways to produce purpose built proxies for human body parts. The ballistic resistance of the skin simulant also became more consistent in its bonded form compared to when it was fitted to a block of gelatine after curing.

By making 4 mm thick gelatine sheets with varying gelatine to water ratios between 30:70% to 45:55%, it is possible to produce skin simulants to a predetermined ballistic resistance that falls within the range of human skin [149]. This could be of value when performing lethality tests for a specific part of the human body.

As many as eleven different natural and synthetic skin simulants, either described earlier in the literature or new materials, were tested on the macroscopics comparability of bullet entry wounds [150]. Dental silicones performed best in that respect.

6.1.2. Impact behavior and effect

In an experimental study with 9 mm Luger bullets fired at human long bones (femurs and humeri), a consistent comminution with a stellate pattern was found [151]. The results indicate that some of the fracture traits depend on bone properties such as shaft diameter, bone length and cortical thickness.

In another study on long bones, porcine femurs were perforated with hollow-point bullets [152]. With some of these bullets, the front part of the jacket was removed. In general, bone comminution and energy transfer was higher when the front part of the jacket was removed.

In a study with flat bones (porcine scapulae), more radiating cracks, crack branching points and longer fracture lengths were found with high energy bullets than with low energy bullets [153].

When 6 mm ball bearing balls were fired through porcine scapulae, two forms of plastic deformation were observed with an SEM in the cortical bone surrounding the entry fracture [154].

When an artificial bone plate is perforated at an angle by a bullet, the

larger fragments are projected from the backside orthogonal from the plate, like glass from a window [155]. The smaller fragments followed the bullet. Most fragments did not produce separate wound tracks in the soft tissue simulant behind the plate but were found in the wound track produced by the bullet.

Experiments were performed with porcine parenchymatous organs imbedded in gelatine that were perforated by 9 mm Luger bullets [156]. Stellate tears, resembling those seen in skin wounds, radiate from the wound tracks in the livers, spleens and kidneys. The difference with other tissues, such as lung tissue that showed tube-shaped wound tracks without cracks, can be explained by the viscoelastic tissue behavior.

Energy transfer to gelatine blocks increased with increasing impact velocity in a study, but this did not have an influence on bullet deflection [157]. The deflection of 9 mm Luger full metal jacket round-nose bullets was a consistent 5.7° to 7.4° for 136 out of the 140 fired shots. The other 4 shots were considered outliers, with values below 5.7°.

Energy transfer of 9 mm nominal caliber bullets was found to increase linear with increasing gelatine thickness from 2 to 12 cm [158]. Energy deposition was calculated based on the loss of velocity when perforating gelatine blocks. An increased loss of energy was observed for higher impact velocities.

The terminal ballistics and flight behavior of 9 mm Luger solid brass hollow point bullets were studied [159]. For elevation angles below 66° bullets descended in a nose-down orientation, while they descended in a base-down orientation for elevation angles over 66°. Energy densities of the descending bullets were calculated. Based on these calculations, bullets fired at elevation angles below 66° seem to be able to perforate unprotected human skin, while those fired at an angle over 66° are unlikely to do so.

Jacketed hollow point bullets (JHP) do not always expand upon penetrating human tissue. This is often attributed to factors such as damage to the bullet's cavity, cavity obstruction by intermediate targets, velocity loss and destabilization. In their study, Paolucci and Diaczuk did not find the latter two to be contributing factors [160]. They did however observe expanding failures for damaged bullet cavities and for obstructed bullet cavities. For obstructed bullet cavities they show that this mostly occurred when the intermediated material was shored against ballistic gelatine. With an intermediate flight of 7 ft (± 2.1 m) between intermediate target and gelatine, the bullets did expand. Such a finding could be helpful when reconstructing accounts of a victim's proximity to for example a perforated wall.

A study looked into the behavior of .22 Long Rifle and .38 Special bullets in ballistic gelatine [161]. The study confirms common knowledge that expanding bullets will penetrate less and that bullets fired with a rifle lead to higher velocities and related kinetic energies.

6.1.3. Lethality

Three papers were found that deal with 'borderline' questions concerning lethality and injury potential of firearms and related items. One study explored, by calculation and experimentation, at what ranges the perforation of human skin is still possible with #4, #6 and #8 lead shot [162]. Pellets of all three sizes penetrated the 1 mm chamois skin simulant at 50 yard. The #4 and #6 size pellets also consistently penetrated the skin simulant at 75 yard, while the #8 size pellets consistently bounced off at this distance. All pellets bounced off at 100 yards, with individual penetrations for a #4 pellet and a #6 size pellet.

Another study explored whether and up to what distance 9 mm P A. K. rubber ball projectiles (8.6 mm, weighing 0.63 g) could perforate human skin [163]. This was possible up to 110 cm. With contact shots, cranial (porcine) bone was perforated.

The last study explored the injury potential of three types of projectile firing objects, generally sold as toys [164]. NERF-guns, gel blasters and airsoft guns were tested against the Australian and New-Zealand threshold for toys of 2500 J/m². The difference between the three classes was large, with only NERF-guns consistently below this value.

Some Eastern European countries allow “non-lethal” or “less-lethal” weapons such as traumatic handguns utilizing rubber bullets. While designed to be non-lethal, casework shows that adding powder to the cartridges can result in higher velocity and specific energy of the rubber bullets. The cartridges can also be modified to propel metal projectiles instead of rubber bullets and the handguns themselves can be converted to a similar design as conventional firearms. All these changes result in a potentially lethal handgun-ammunition combination [165].

6.1.4. Miscellaneous

Experiments were performed to study how much the positions of entrance and exit wounds shift when the body of a victim is scanned in another position than it had when hit [166]. The recorded values can be used to estimate errors when the position of an entrance and exit wound is used to indicate a bullet trajectory through a body in reconstructions.

An impact of a .40 S&W bullet to a skull filled with neural tissue was simulated with a finite element analysis [167]. The study was intended as a proof of principal and no comparison to experimental results were included.

A roughly comparable approach, but with a comparison to a high-profile case was made in another study [168]. The effect of a 6.5 mm Carcano bullet on the head of J.F. Kennedy was simulated from several different angles. The results of the simulations were compared to the known results from the autopsy findings. The wounds were most consistent with a shot from behind (in line with the conclusion from the original Warren commission).

GSR on charred (cremated) bone could still be detected with XRF spectroscopy in an experimental setting [169]. Full metal jacket and lead bullets were used. More GSR was found after impact from lead bullets.

The sizes of 9 mm Luger bullet holes in the skin of anterior and posterior porcine trunks were compared [170]. The maximum diameter and the area of each skin defect was measured with an image analysis system. Defects in the anterior skin were smaller than in the posterior skin. This was attributed to the thinner dermis in the latter.

Three types of rifle bullets were fired in gelatine and the distribution of bullet fragments in the wound track was compared with computed tomography [171]. Fragmentation was different for all three types. Surprisingly, the tested full metal jacket type fragmented as well.

Winchester 115 gr. 9 mm Service Grade Ammunition with a truncated cone shape do not yaw and produce straight wound tracks. As a result, it may be hard to differentiate between the entrance and exit hole in human bodies [172].

An atypical entry wound pattern was observed [173]. Experimentation revealed that the pattern was caused by the ported barrel of a Taurus Tracker model 627 model revolver.

A total of 251 cranial traumas caused by 9 mm Luger FMJ bullets in the remains of 161 executed individuals were studied [174]. The occurrence of Circumferential Delamination Defects (CDD) (chipping of the outer layer of cortical bone around the entrance defect) was found to be an indicator of short-range or contact shots.

6.2. Case reports

6.2.1. Atypical suicides

Six case reports were found describing suicides that could have been mistaken for homicides. In one of these reports, five fatalities with self-inflicted gunshot wounds to the posterior head and neck of adult males were described [175]. The indications that these wounds were self-inflicted rather than resulting from homicides are discussed.

The other five papers emphasize that the occurrence of multiple gunshot wounds in a single victim can be observed with suicides. A case was reported where a victim first shot himself in the chest and neck before inflicting a final and fatal shot to the central forehead [176]. The first two wounds seriously injured the victim but still allowed him to manually cycle the 16 gauge repeating shotgun twice before inflicting

the third shot. A somewhat similar case was described in which the victim shot himself twice in the chest and managed to reload the double-barrel 12 gauge shotgun to fire a third and final shot to the chest [177].

The next case describes a suicide with a 6.35 mm (.25 Auto) caliber pistol [178]. The victim was found dead with four self-inflicted gunshot wounds to the chest. According to witness statements, the time between the first and last shot might have been 2 min.

Another study starts with an overview of earlier work, from which the authors derive that between 0 and 5% of all suicides with firearms include more than one shot [179]. The authors add to this body of knowledge by describing three more cases, all with multiple shots to the chest. In two of these incidents, two shots were fired with a .38 Special revolver and two with a .32 Auto pistol. In the third incident, three shots were fired with .38 Special cartridges from a .357 Magnum revolver.

The last paper covers a dramatic fatality with a pneumatic nail gun [180]. This is a tool rather than a firearm and the authors describe the wounds as sharp force trauma instead of gunshot injuries. Because the wounds were inflicted in a similar manner as with a firearm and because the paper was written with the same intent as the aforementioned papers, it is included in this review. The victim was found with as many as 21 nails in the left and right temple region. The conclusion was that the wounds were self-inflicted.

6.2.2. Atypical entrance wounds and bullet behavior

Six reports were found with atypical wounds from firearms that have the potential to be misleading in forensic casework. Flat-point full metal jacket bullets can perforate one and even two victims without notable deflection or yaw [181]. The entrance wounds in the second and possibly even the third victim are round, without any indication of the instability that is expected with most other bullet types after perforating a victim.

Solid copper bullets from Lehigh Defense Xtreme Defense Penetrator Ammunition produce entrance wounds mimicking stab wounds resulting from Philips or Robertson screwdrivers [182,183].

A .32 Auto full metal jacket bullet followed an atypical, curved trajectory through the subcutaneous soft tissue of a female victim [133, 134]. Contact with bone could be excluded as a reason for bullet deflection.

A contact wound from a Frommer Stop semi-automatic pistol, with its recoil spring above the barrel, resembles the wounds produced by most other semi-automatic pistols after 180° rotation [184].

The term “bullet track skin defect” was proposed for a defect caused by a bullet traveling within the body and damaging the skin from below (typically tangentially) without exiting the body [185]. The appearance of these defects varies in shape and size. Examples from eight different cases are presented, all with full- or partial thickness skin defects.

Two fragments from a single semi-jacketed .223 Remington bullet killed a victim in a residence. The bullet originated from a nearby outdoor shooting incident. The bullet fragmented when it passed through the residence outer wall [186]. The fragments and entry wounds are described and discussed.

6.2.3. Miscellaneous

A man died as a result of a shot with a PIEXON JPX4 hand weapon [187]. This weapon is a four-barreled derringer type pistol that projects a jet of irritant gel (containing capsaicin) by means of a pyrotechnic delivery system. Subsequent experiments showed that the jet can rupture a chamois leather skin simulant up to 50 cm from the muzzle. Fissures were observed in a supporting block of synthetic gel under the skin simulant up to 30 cm from the muzzle.

A man died as a result of an occupational accident in a marble quarry [188]. An entrance wound, resembling an angled gunshot wound, was observed between the neck and the left shoulder. A 3.5 cm long metal projectile, originating from a diamond wire saw machine, was found in the chest cavity and identified as the cause of death.

A 0.7 kg piece of stucco apparently caused a deep dent in a wooden front door after it was propelled by a lawn mower over a distance of some 60 m from the respective residence. The case was examined by a firearm examiner, after a first expert excluded the lawn mower as being the possible cause. The examination consisted of calculating the possible trajectory of the piece of stucco and discussing possible other mechanisms that could have propelled it [189].

The last paper discusses the incorrect use of the term “cavitation” for the formation of a temporary cavity by a projectile in soft tissue. The difference between temporary cavity formation and the actual meaning of the word cavitation in a hydrodynamic context is explained. Examples are given of where this confusion of definitions can be harmful [190].

7. Meetings

A concise overview of the 27th (online) annual Firearm/Gunshot Residue export working group of the European Network of Forensic Science Institutes (ENFSI) is published [191]. The authors provide a description of the working group and announcement made during the meeting. In addition they provide a summary of the various presentations which are of interest to forensic examiners worldwide.

A concise overview of the 23rd Triennial meeting of the International Association of Forensic Science (IAFS) held in 2023 in Sydney, Australia is published [192]. The authors provide a description of the various presentations relating to firearm and toolmark disciplines.

A concise overview of the technical presentations delivered on the 13th of September 2022 during the 25th Australian and New Zealand Forensic Science Society Symposium (ANZFSS) and relating to the firearms and toolmark discipline is published [193].

A number of conference abstracts are published by Forensic Science International: Synergy. For firearm examiners the following may be of interest:

- A workshop was given on congruent matching methods which divide toolmarks into correlation cells to allow for pairwise cell comparison instead of comparing entire images. These methods include congruent matching cells, congruent matching features and congruent matching profile segments. With the help of statistics a weight of evidence can be calculated for a comparison. NIST is working on a commercial software package to support examiners [194].
- A presentation on the congruent matching methods developed by NIST to explain how they work and present their result. In addition, they explain the use of the NIST Ballistic Database Testbed and state that the developed methodology could be valuable for fast searches in large ballistic databases including LR calculations [195].
- A workshop was given with the aim to provide forensic science managers with an overview of new technologies to facilitate firearm comparisons with computer-based methods using 3D measurements of the features in bullets and cartridge cases. They also showed to use of virtual comparison microscopy [196].
- The uptake of virtual comparison microscopy and its benefits are explained. The software allows examiners to compare features that were previously measured in 3D. The software allows for action such as recording notes and conclusions, exports to LIMS, blind peer review (see also [197] for additional information of the added value of this) and highlighting. The presenter states it helps to improve turn-around-times and to reduce backlogs [198].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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